More Electrode Potentials

1 Aluminium—oxygen cells are being investigated for powering vehicles.

The **reactions** at each electrode are shown below.

X 4 $Al(s) + 4OH^-(aq) \rightarrow Al(OH)_4^-(aq) + 3e^-$

X 3 $O_2(g) + 2H_2O(l) + 4e^- \rightarrow 4OH^-(aq)$

+0.4

(i) The standard electrode potential for the O_2 / OH^- redox system is +0.40 V.
The standard cell potential of an aluminium—oxygen cell is 2.71 V.

What is the standard electrode potential of the aluminium redox system in this cell?

$2.71 = 0.4 - x$

$x = -2.31$

standard electrode potential = -2.31 V [1]

(ii) Construct the overall cell equation for an aluminium—oxygen cell.

$4Al + 6OH^-$

[2]

2 A standard cell is set up from $H^+(aq)|H_2(g)$ and $Ag^+(aq)|Ag(s)$ half cells.

The standard electrode potentials of the redox systems are shown below.

$$H^+(aq) + e^- \rightleftharpoons \tfrac{1}{2}H_2(g) \qquad E^\ominus = +0.00\ V$$

$$Ag^+(aq) + e^- \rightleftharpoons Ag(s) \qquad E^\ominus = +0.80\ V$$

What is the reaction at the negative electrode of the cell?

A $H^+(aq) + e^- \rightarrow \tfrac{1}{2}H_2(g)$
B $\tfrac{1}{2}H_2(g) \rightarrow H^+(aq) + e^-$
C $Ag^+(aq) + e^- \rightarrow Ag(s)$
D $Ag(s) \rightarrow Ag^+(aq) + e^-$

Your answer B

[1]

3 This question is about some reactions of d block elements and their ions.

Table 21.1 shows standard electrode potentials which will be needed within this question.

$Zn^{2+}(aq) + 2e^-$	$\rightleftharpoons$	$Zn(s)$	$E^\ominus = -0.76\,V$
$Cr^{3+}(aq) + e^-$	$\rightleftharpoons$	$Cr^{2+}(aq)$	$E^\ominus = -0.42\,V$
$Ni^{2+}(aq) + 2e^-$	$\rightleftharpoons$	$Ni(s)$	$E^\ominus = -0.25\,V$
$I_2(aq) + 2e^-$	$\rightleftharpoons$	$2I^-(aq)$	$E^\ominus = +0.54\,V$
$Fe^{3+}(aq) + e^-$	$\rightleftharpoons$	$Fe^{2+}(aq)$	$E^\ominus = +0.77\,V$
$Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e^-$	$\rightleftharpoons$	$2Cr^{3+}(aq) + 7H_2O(l)$	$E^\ominus = +1.33\,V$
$H_2O_2(aq) + 2H^+(aq) + 2e^-$	$\rightleftharpoons$	$2H_2O(l)$	$E^\ominus = +1.78\,V$

Table 21.1

Use the information in **Table 21.1** to help you answer both parts of this question.

(i) Write the overall equation for the oxidation of Fe^{2+} by acidified H_2O_2.

$H_2O_2 + 2H^+ + 2Fe^{2+} \rightarrow 2H_2O + 2Fe^{3+}$ [1]

(ii) Zinc reacts with acidified $Cr_2O_7^{2-}$ ions to form Cr^{2+} ions in two stages.

Explain why this happens in terms of electrode potentials and equilibria.

Include overall equations for the reactions which occur.

$Zn^{2+} + 2e^- \rightleftharpoons Zn \quad E = -0.76$

$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightleftharpoons 2Cr^{3+} + 7H_2O \quad E = 1.33$

$E_{cell} = 1.33 - -0.76 = 2.09$

---[4]

4 Standard electrode potentials for four redox systems are shown in **Table 19.1**.

Redox system	Half-equation	$E^{\ominus}$ / V
1	$CO_2(g) + 2H^+(aq) + 2e^- \rightleftharpoons HCOOH(aq)$	–0.11
2	$HCOOH(aq) + 2H^+(aq) + 2e^- \rightleftharpoons HCHO(aq) + H2O(l)$	–0.03
3	$Ag^+(aq) + e^- \rightleftharpoons Ag(s)$	+0.80
4	$MnO_4^-(aq) + 8H^+(aq) + 5e^- \rightleftharpoons Mn^{2+}(aq) + 4H_2O(l)$	+1.51

Table 19.1

A student warms a mixture of methanal, HCHO, and acidified potassium manganate(VII).

The student observes gas bubbles.

Explain this observation in terms of electrode potentials and equilibria.

Include overall equations in your answer.

[4]

5 Four redox systems relevant to hydrogen–oxygen fuel cells are shown below.

	$E^\ominus$ / V
$H_2O(l) + e^- \rightleftharpoons OH^-(aq) + \frac{1}{2}H_2(g)$	–0.83
$H^+(aq) + e^- \rightleftharpoons \frac{1}{2}H_2(g)$	0.00
$\frac{1}{2}O_2(g) + H_2O(l) + 2e^- \rightleftharpoons 2OH^-(aq)$	+0.40
$\frac{1}{2}O_2(g) + 2H^+(aq) + 2e^- \rightleftharpoons H_2O(l)$	+1.23

Which statement(s) is/are correct for an alkaline hydrogen–oxygen fuel cell?

1 The reaction at the positive electrode is: $\frac{1}{2}O_2(g) + 2H^+(aq) + 2e^- \rightarrow H_2O(l)$

2 The overall cell reaction is: $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$.

3 The cell potential is 1.23 V.

A 1, 2 and 3

B Only 1 and 2

C Only 2 and 3

D Only 1

Your answer C **[1]**

6 The redox equilibria for a hydrogen–oxygen fuel cell in alkaline solution are shown below.

$2H_2O(l) + 2e^- \rightleftharpoons H_2(g) + 2OH^-(aq)$ $\quad E^\ominus = -0.83$ V

$^1/_2O_2(g) + H_2O(l) + 2e^- \rightleftharpoons 2OH^-(aq)$ $\quad E^\ominus = +0.40$ V

What is the equation for the overall cell reaction?

A $H_2(g) + 4OH^-(aq) \rightarrow 3H_2O(l) + {}^1/_2O_2(g)$

B $3H_2O(l) + {}^1/_2O_2 \rightarrow H_2(g) + 4OH^-(aq)$

C $H_2O(l) \rightarrow H_2(g) + {}^1/_2O_2(g)$

D $H_2(g) + {}^1/_2O_2(g) \rightarrow H_2O(l)$

Your answer D **[1]**

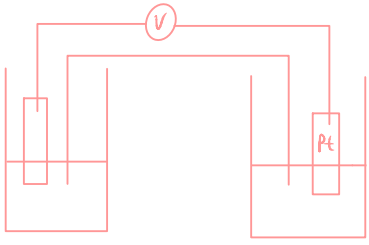
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3	$Ag^+(aq) + e^- \rightleftharpoons Ag(s)$	+0.80
4	$MnO_4^-(aq) + 8H^+(aq) + 5e^- \rightleftharpoons Mn^{2+}(aq) + 4H_2O(l)$	+1.51

Table 19.1

A student sets up a standard cell in the laboratory based on redox systems **3** and **4**.

Draw a labelled diagram to show how this cell could be set up to measure its standard cell potential at 298 K.



[3]

END OF QUESTION PAPER

Y1 Titration 1

1(a) A student carries out a titration to determine the molar mass and structure of a weak acid **A**.

The student follows the method below.

- Dissolve a weighed mass of **A** in 100 cm^3 of distilled water and make the solution up to 250 cm^3 in a beaker.
- Add the solution of **A** to a burette.
- Titrate the solution of **A** with a standard solution of sodium hydroxide, NaOH.

What is meant by the term standard solution?

known concentration and a fixed volume [1]

(b) Sodium hydroxide is an alkali.

What is meant by the term alkali?

base dissolved in water releases OH^- ions

[1]

2(a) A student carries out a titration to determine the concentration of some hydrochloric acid.

The student titrates the hydrochloric acid against a standard solution of sodium carbonate, Na_2CO_3. The equation is shown below.

$$Na_2CO_3(aq) + 2HCl(aq) \rightarrow 2NaCl(aq) + H_2O(l) + CO_2(g)$$

- The student prepares 0.150 mol dm^{-3} Na_2CO_3 in a 250.0 cm^3 volumetric flask.
- The hydrochloric acid is added to a 50.0 cm^3 burette.
- The student pipettes the $Na_2CO_3(aq)$ using a 25.0 cm^3 pipette.

The student's burette readings are shown in the table.
The rough titre has been omitted.

(i) Complete the table by adding the titres to the table.

Final reading / cm^3	24.60	48.45	34.30
Initial reading / cm^3	0.40	24.60	10.00
Titre / cm^3	24.20	23.85	24.30

[1]

(ii) Calculate the mean titre of HCl, to the nearest 0.05 cm^3, that the student should use for analysing the results.

mean titre = ... cm^3 **[1]**

(b) Calculate the concentration, in mol dm^{-3}, of the hydrochloric acid.

Give your answer to **3** significant figures.

concentration of HCl = ... mol dm^{-3} **[3]**

(c) In the titrations, the student measured volumes with a pipette and a burette.

- The pipette had an uncertainty of ± 0.04 cm^3 in the volume measured.
- The burette had an uncertainty of ± 0.05 cm^3 in the volume measured.

Determine whether the volume measured by the pipette or the volume measured by the burette has the greater percentage uncertainty.

Error = $\frac{\text{amount} \times \text{value}}{\text{quantity}}$

[2]

3 This question is about barium hydroxide.

A student carries out a titration to determine the concentration of an aqueous solution of $Ba(OH)_2$.

The student adds 25.0 cm^3 of the $Ba(OH)_2(aq)$ solution to a conical flask.
The student titrates this solution by adding 0.160 mol dm^{-3} $HNO_3(aq)$ from the burette.

The equation is shown below.

$Ba(OH)_2(aq) + 2HNO_3(aq) \rightarrow Ba(NO_3)_2(aq) + 2H_2O(l)$

The student repeats the titration until concordant titres are obtained.

The mean titre of 0.160 mol dm^{-3} $HNO_3(aq)$ is 26.75 cm^3.

(i) What is meant by concordant titres?

results within 0.1 of each other

[1]

(ii) Calculate the concentration, in mol dm^{-3}, of the $Ba(OH)_2(aq)$ solution.

	$Ba(OH)_2$	
moles		
conc		
volume		

concentration of $Ba(OH)_2(aq)$ = ... mol dm^{-3} **[3]**